

Error correction technique:- Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is moved or stored from the sender to the receiver. It is technique developed by R.W. Hamming for error correction.

the first step is to identify the bit position of the data & all the bit positions which are powers of 2 are marked as parity bits (e.g. 1, 2, 4, 8, etc.). The following image will help in visualizing the received hamming code of 7 bits.

Step 1: Find the number of parity bits

Use the formula:

$$2^r \geq m+r+1$$

Where:

- **m = 4** (data bits)
- **r = parity bits**

Try **r = 3**:

$$2^3 = 8 \geq 4 + 3 + 1 = 8$$

So, **3 parity bits** are needed.

P1 checks all positions whose last (rightmost) binary bit is 1.

001 ✓ (1)

010 ✗ (2)

011 ✓ (3)

100 ✗ (4)

101 ✓ (5)

110 ✗ (6)

111 ✓ (7)

P2 checks all positions whose middle (second) bit is 1.

001 ✗ (1)

010 ✓ (2)

011 ✓ (3)

100 ✗ (4)

101 ✗ (5)

110 ✓ (6)

111 ✓ (7)

P4 → Checks positions where the leftmost (third) bit = 1.

001 ✗ (1)

010 ✗ (2)

011 ✗ (3)

100 ✓ (4)

101 ✓ (5)

110 ✓ (6)

111 ✓ (7)

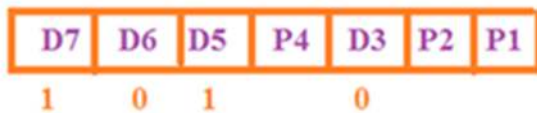
P8 checks all positions where the 8's bit (leftmost bit) is 1.

Position	Binary (4-bit)	8 bit
8	1000	1 ✓
9	1001	1 ✓
10	1010	1 ✓
11	1011	1 ✓
12	1100	1 ✓
13	1101	1 ✓
14	1110	1 ✓
15	1111	1 ✓

$P1=(p1,d3,d5,d6)=(1,3,5,7)$
 $P2=(p2,d3,d6,d7)=(2,3,6,7)$
 $P4=(p4,d5,d6,d7)=(4,5,6,7)$

7-bit Hamming code which is 1010011.
even parity Hamming code.

find parity bits of data 1010



$P1=(,3,5,7)= (P1,0,1,1)= P1=0$
 $P2=(,3,6,7)= (P2,0,0,1)= P2=1$
 $P4=(,5,6,7)= (P4,1,0,1)= P4=0$

the hamming code is



Let Detect and correct Hamming code
1011010



$p1=(p1,3,5,7)= (0,1,1)= 0$
 $p2=(p2,3,6,7)= (0,0,1)= 1$
 $p4=(p4,5,6,7)= (1,0,1)= 0$

Error Word



correct data will be:



Q. Find the parity bit of 8 bits data 10110101 hamming code

Find r such that:

$$2^r \geq m+r+1 = 2^4 = 16 \geq 8+4+1=13$$

So, **4 parity bits** are required.

Parity bits: **P1, P2, P4, P8**

D8	D7	D6	D5	P8	D4	D3	D2	P4	D1	P2	P1
1	0	1	1	?	0	1	0	?	1	?	?

Position Binary Last bit P1 checks?

1	0001	1	✓
2	0010	0	✗
3	0011	1	✓

Position Binary Last bit P1 checks?

4	0100	0	x
5	0101	1	✓
6	0110	0	x
7	0111	1	✓
8	1000	0	x
9	1001	1	✓
10	1010	0	x
11	1011	1	✓
12	1100	0	x

P1 checks positions 3, 5, 7, 9, 11

P2 checks positions 3, 6, 7, 10, 11

P4 checks positions 5, 6, 7, 12

P8 checks positions 9, 10, 11, 12

Do Yourself